# EXPERIMENT – 1

## Installation and Configuration of Kali Linux in Oracle VirtualBox

### Aim

To install and configure Kali Linux as a virtual machine using Oracle VirtualBox for performing network security and ethical hacking experiments.

### Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- System Requirements:
    - RAM: Minimum 4 GB (8 GB recommended)
    - Disk Space: Minimum 30 GB
    - Processor: Intel/AMD with Virtualization support

### Introduction:

Kali Linux is a Debian-based Linux distribution developed for penetration testing, digital forensics, and cybersecurity analysis.
Installing Kali Linux inside Oracle VirtualBox provides a safe and isolated environment to practice security experiments without affecting the host system.

VirtualBox is an open-source virtualization tool that allows running multiple operating systems simultaneously on a single physical machine.

### Procedure

Step 1: Enable Virtualization

1. Restart the system and enter BIOS/UEFI settings.
2. Enable Intel VT-x / AMD-V.
3. Save changes and reboot the system.

Step 2: Install Oracle VirtualBox

1. Download Oracle VirtualBox from the official website.
2. Run the installer and follow default installation steps.
3. Complete the installation and restart the system if required.

Step 3: Download Kali Linux

1. Visit the official Kali Linux website.
2. Download the Kali Linux Installer ISO (64-bit).

Step 4: Create a New Virtual Machine

1. Open Oracle VirtualBox.
2. Click New.
3. Enter:
    - Name: Kali Linux
    - Type: Linux
    - Version: Debian (64-bit)
4. Click Next.

Step 5: Allocate Memory and CPU

- Memory Size: 2048 MB or higher
- Processor: 2 CPUs
- Click Next.

Step 6: Create Virtual Hard Disk

1. Select Create a virtual hard disk now.
2. Choose VDI (VirtualBox Disk Image).
3. Select Dynamically allocated.
4. Set disk size to 30 GB.
5. Click Create.

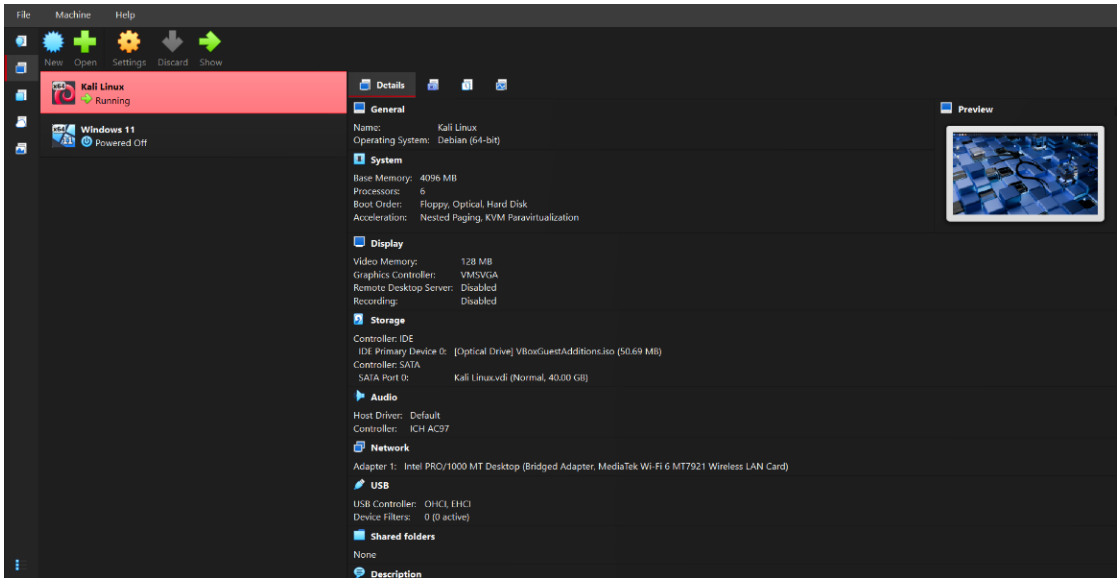
Step 7: Attach Kali Linux ISO

1. Select the created VM → Settings.
2. Go to Storage.
3. Under Controller IDE, select Empty.
4. Click Choose a disk file and select Kali Linux ISO.
5. Click OK.

Step 8: Install Kali Linux

1. Start the virtual machine.
2. Select Graphical Install.
3. Choose:
    - Language
    - Location
    - Keyboard layout
4. Configure network automatically.
5. Set:
    - Username
    - Password
6. Choose:
    - Guided – Use entire disk
    - All files in one partition
7. Confirm disk changes and continue installation.
8. After completion, reboot the system.

**Output:**


File
Machine
Help
New
Open
Settings
Discard
Show
Kali Linux
Running
Windows 11
Powered Off
Details
General
Name: Kali Linux
Operating System: Debian (64-bit)
System
Base Memory: 4096 MB
Processors: 6
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, KVM Paravirtualization
Display
Video Memory: 128 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Primary Device 0: [Optical Drive] VBoxGuestAdditions.iso (50.69 MB)
Controller: SATA
SATA Port 0: Kali Linux.vdi (Normal, 40.00 GB)
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (Bridged Adapter, MediaTek Wi-Fi 6 MT7921 Wireless LAN Card)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
Preview

Applications
Home
File System
Trash
19:02

```
┌──(dymonish㉿Kali)-[~]
└─$ uname -a
Linux Kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64 GNU/Linux

┌──(dymonish㉿Kali)-[~]
└─$
```

```
┌──(dymonish㉿Kali)-[~]
└─$ sudo apt upgrade
The following packages were automatically installed and are no longer required:
  bloodhound.py  libavformat61          libgav1-1          libmplex2-2.1-0t64  libsphinxbase3t64  mesa-vdpau-drivers
  curlftpfs      libconfig-inifiles-perl libmjpegutils-2.1-0t64 libpocketsphinx3    libswscale8        pocketsphinx-en-us
  libavfilter10  libfuse2t64            libmpeg2encpp-2.1-0t64 libpostproc58       libvdpau-va-gl1    vdpau-driver-all
Use 'sudo apt autoremove' to remove them.

Upgrading:
  7zip                      libdrm-intel1             libsodium23                 python3-gi-cairo
  adduser                   libdrm-nouveau2           libsoup-3.0-0               python3-greenlet
  alsa-ucm-conf             libdrm-radeon1            libsoup-3.0-common          python3-gssapi
  amd64-microcode           libdrm2                   libsoxr0                    python3-gvm
  apache2                   libdvdnav4                libspa-0.2-bluetooth        python3-hamcrest
  apache2-bin               libdvdread8t64            libspa-0.2-modules          python3-html5lib
  apache2-data              libebackend-1.2-11t64     libspeechd-module0          python3-impacket
  apache2-utils             libebook-1.2-21t64        libspeechd2                 python3-iniconfig
  apparmor                  libebook-contacts-1.2-4t64 libsqlcipher2              python3-ipwhois
  apt                       libedata-book-1.2-27t64   libss2                      python3-jeepney
  apt-utils                 libedataserver-1.2-27t64  libstrongswan              python3-joblib
  arping                    libedit2                  libstrongswan-standard-plugins python3-jq
  at-spi2-common            libegl1                   libsvn1                     python3-jsonschema-specifications
  at-spi2-core              libevdev2                 libswresample5              python3-jwt
  binutils                  libext2fs2t64             libsyn123-0t64              python3-kiwisolver
  binutils-common           libfdisk1                 libsynctex2                 python3-kombu
  binutils-mingw-w64-i686   libffado2                 libsystemd-shared           python3-ldb
  binutils-mingw-w64-x86-64 libffi-dev                libsystemd0                 python3-legacy-cgi
  binutils-x86-64-linux-gnu libffi8                   libtalloc2                  python3-legacycrypt
  bsdextrautils             libfile-desktopentry-perl libtbb12                    python3-loguru
  bsdutils                  libfluidsynth3            libtbbbind-2-5              python3-louis
  burpsuite                 libfreerdp-client3-3      libtbbmalloc2               python3-lxml
  cadaver                   libfreerdp3-3             libtdb1                     python3-lz4
  certipy-ad                libfreetype6              libteckit0                  python3-macholib
  chromium                  libfribidi0               libtevent0t64               python3-markdown
```

```
Installing dependencies:
  binutils-mingw-w64-base  lib32stdc++6              libclang-common-21-dev  libllvm21           libswscale9                    llvm-21-tools
  clang-21                 libavcodec62              libclang-cpp21          libmjpegutils-2.2-0 linux-image-6.18.3+kali+1-amd64 systemd-userdbd
  clang-tools-21           libavfilter11             libclang-rt-21-dev      libmpeg2encpp-2.2-0 llvm-21
  ibus-gtk3                libavformat62             libclang1-21            libmplex2-2.2-0     llvm-21-dev
  ibus-gtk4                libavutil60               libgav1-2               libsimdutf29        llvm-21-linker-tools
  lib32gcc-s1              libc-gconv-modules-extra  libibus-1.0-5          libswresample6      llvm-21-runtime

Suggested packages:
  clang-21-doc  wasi-libc  libcuda1  libnvcuvid1  libnvidia-encode1  debian-kernel-handbook  linux-doc-6.18  llvm-21-doc

Not upgrading:
  libavfilter10  libavutil59   libgbm1          libglx-mesa0  mesa-libgallium  python3-tables-lib
  libavformat61  libegl-mesa0  libgl1-mesa-dri  libswscale8   nodejs

Summary:
  Upgrading: 948, Installing: 32, Removing: 0, Not Upgrading: 11
  Download size: 2,613 MB
  Space needed: 1,438 MB / 21.8 GB available

Continue? [Y/n] y
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 libc-gconv-modules-extra amd64 2.42-5
Ign:2 http://http.kali.org/kali kali-rolling/main amd64 libc-l10n all 2.42-5
Ign:3 http://http.kali.org/kali kali-rolling/main amd64 locales all 2.42-5
Ign:4 http://http.kali.org/kali kali-rolling/main amd64 libc6 amd64 2.42-5
Ign:5 http://http.kali.org/kali kali-rolling/main amd64 libc-bin amd64 2.42-5
Ign:6 http://http.kali.org/kali kali-rolling/main amd64 libc-dev-bin amd64 2.42-5
Ign:7 http://http.kali.org/kali kali-rolling/main amd64 libc6-dev amd64 2.42-5
Ign:8 http://http.kali.org/kali kali-rolling/main amd64 libc6-i386 amd64 2.42-5
Ign:9 http://http.kali.org/kali kali-rolling/main amd64 linux-libc-dev all 6.18.3-1kali2
Ign:10 http://http.kali.org/kali kali-rolling/main amd64 libncurses-dev amd64 6.6+20251231-1
Ign:11 http://http.kali.org/kali kali-rolling/main amd64 libncurses6 amd64 6.6+20251231-1
Ign:11 http://http.kali.org/kali kali-rolling/main amd64 libncurses6 amd64 6.6+20251231-1
```

**Result**

Kali Linux was successfully installed and configured in Oracle VirtualBox.
The virtual machine is ready for performing network security and ethical hacking experiments.